Car Rental Company

Initial Project Report

Team members:

Tymoteusz Kłoska

Szymon Molicki

Marcjan Zawadzki

Mateusz Znaleźniak

Filipa Delfino

Academic year: 2025/2026

Section: Informatics, Section 6

Supervisor: Hafedh Zghidi

Date: April 8, 2026

Project Overview

The goal of the project is to design and implement a system for managing a car rental company. The application should support customer registration and profile editing, browsing available cars, making rental reservations, reviewing rental history, generating reports, online payments and using discount coupons. On the company side, the system should support fleet management, customer management, rental settlement, invoice generation and informing regular customers about promotions and discounts.

Selected Tech Stack

The project uses a Python-based stack. The selected technologies are presented in Table 1.

Layer	**Technology**	**Purpose**
Frontend	React	User interface for customers and rental company staff.
Backend API	Python + FastAPI	REST API, business logic, validation and documentation.
ORM	SQLAlchemy	Object-relational mapping between Python models and the database.
Migrations	Alembic	Versioned database schema changes.
Database	PostgreSQL	Persistent relational data storage.
Containers	Docker Compose	Local development environment and service orchestration.

Table 1: Selected project technology stack.

Initial Functional Scope

Actor	**Key functions**
Customer	Register and edit personal data, browse available vehicles, create reservations, review rental history, pay online and use discounts.
Staff	Manage cars and customers, settle rentals, issue invoices and communicate promotions to regular customers.

Table 2: Condensed functional scope.

Database Diagram

The current database design is centered around the `RENTAL` table, which connects customers, cars, pickup and return locations and optional discounts.

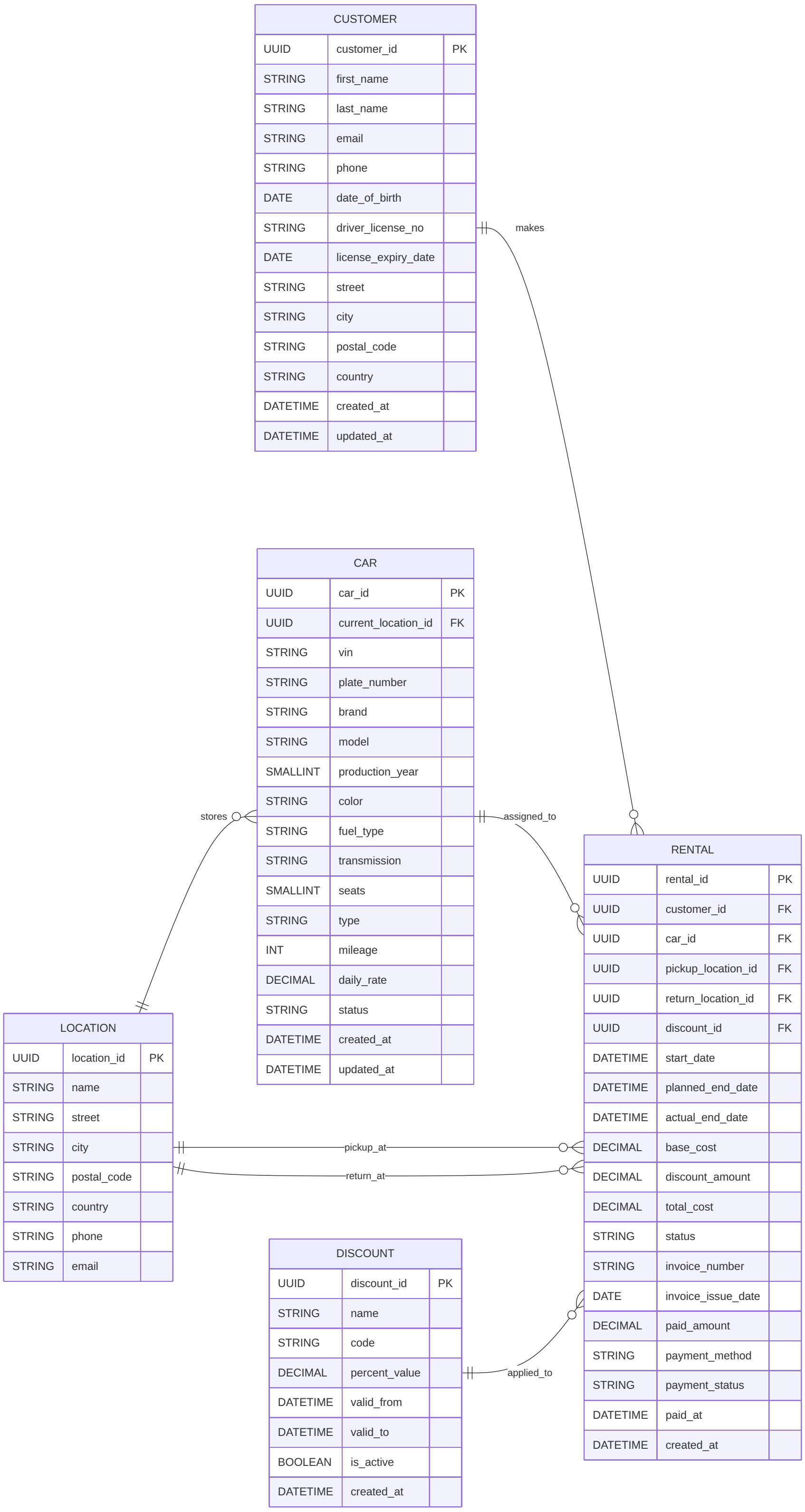

Figure 1: Database diagram in table form.

Core Entities

Entity	Role in the system
`CUSTOMER`	Stores personal, contact, and driving license data required for rentals.
`LOCATION`	Represents rental branches used as car storage points and pickup or return places.
`CAR`	Stores vehicle identity, characteristics, status, rate, mileage and current location.
`DISCOUNT`	Stores discount codes and percentage-based promotions with validity dates.
`RENTAL`	Central transactional entity describing reservation and rental execution, cost calculation, invoice data and payment status.

Table 3: Main entities visible in the current database diagram.

Conclusion

This first report presents the agreed project direction in a concise form: the project scope, the selected Python-based technology stack and the initial relational database design.